

T-OCS V5.0

TIME-PROGRAMMABLE OCCUPANT CONTROL SEAT PATENT DRAFT & INVENTION DISCLOSURE PACKAGE

Vehicle-seat architecture for time-dependent occupant trajectory management, controlled ride-down, progressive energy absorption, multi-state locking, and distributed load paths

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Priority date / filing data: [TO BE SUPPLIED]
Docket: T-OCS-V5-PAT-001
Version: V5.0 patent-preparation draft 2026-09-02

CRITICAL LEGAL / ENGINEERING STATUS

This document is a technical patent-preparation draft and invention disclosure, not a filed patent application and not legal advice. It intentionally distinguishes source-derived facts, engineering proposals, and unvalidated assumptions. No claim of novelty, patentability, safety, regulatory compliance, or injury reduction is made until a qualified patent attorney and independent engineering validation review the work.

Core disclosure thesis: the invention is not a literal local gravity generator. It is a seat system that time-programs the mechanical state of the seat and its force paths so that seat translation, rotation, restraint interaction, compliance and rebound are coordinated during a crash event. The effective-gravity interpretation is an inertial-frame concept, $g_{eff}(t)=g-a_{seat}(t)$, not creation of a new gravitational field.

1. RUTHLESS PATENTABILITY ASSESSMENT

The current concept contains valuable engineering structure, but several broad elements are already crowded by published vehicle-seat patents. A broad claim such as “seat + rail + energy absorber + inertial lock + seatback motion” would be vulnerable. The present draft therefore focuses on combinations and control relationships that are more specific: time-phased state transitions, decoupling of ride-down stroke from seatback rotation control, redundant parallel load paths, explicit rebound state, and coordinated occupant trajectory management.

Feature	Prior-art pressure	Draft strategy
Seat rail + energy attenuation	Very high - known in published seat patents	Do not claim broadly; claim specific multi-state architecture and control relationships.
Inertial lock / cam / pawl	High - classic and modern recliner/rail locks exist	Claim only as part of a coordinated T-OCS state machine / redundant path if supported.
Anti-submarining seat-pan control	High - established field	Use as one element of a larger coordinated load-path system.
Programmable time-phase ride-down	Potentially differentiating	Make timing/state transitions, sensor inputs, and mechanical consequences explicit.
Decoupled rotation vs. translation energy management	Potentially differentiating	Draft independent and dependent claims around separate controlled paths.
Rebound as an explicit controlled phase	Potentially differentiating in combination	Claim dedicated rebound control and lock logic in relation to ride-down.
Front + rear modular seat application	Likely obvious by itself	Keep as system embodiment, not core novelty.

Prior-art warning

Google Patents results include published seat architectures that already describe energy-attenuating seats with track movement, rotary or hydraulic attenuation, and non-linear motion; other patents describe inertial locking and rail energy absorbers. These are not merely theoretical similarities - they overlap important blocks of the present concept. Therefore, the patent position must rely on concrete claim limitations and a documented invention date/priority strategy, not on the label “effective gravity.”

2. GOOGLE PATENTS PRIOR-ART LANDSCAPE

Publication	Subject	Relevant disclosure	Risk to broad claim
US20200231073A1 / US11390198B2	Energy attenuating vehicle seat assembly	Track-based energy attenuation; seat back/pan can move non-linearly; hydraulic/rotary alternatives described.	High overlap with rail + energy attenuation block.
EP3911536A1	Energy attenuating vehicle seat assembly	Rotary energy attenuation cooperating with a track; fluid/actuator/occupant adjustment systems.	High overlap; claim only specific new relationships.
WO2021197809A1	Seat rail pair + energy absorber + locking device	Crash-triggered unlocking of rail coupling, energy absorber in load path, mechanical/electrical release.	High overlap with rail/lock/absorber combination.
US4634182A	Seatback recliner mechanism and inertia operated lock	Pawl, cam, inertial pendulum, crash locking logic.	High overlap with a generic inertial lock.
US6050635A	Seat pan with anti-submarining device	Pelvis anti-submarining cross-member with controlled force / energy dissipation.	High overlap with generic anti-submarine element.
US5125598A	Pivoting energy attenuating seat	Seat pan/energy attenuator arrangement; crash-induced controlled seat movement.	Medium-high overlap.

Design-around / differentiation candidates

- A mechanical state machine in which a first state preserves ordinary seat adjustment, a second state arms an inertial or sensed trigger, a third state releases controlled ride-down, a fourth state controls rebound, and a final state secures the structure after the event.
- Two parallel longitudinal load paths where the energy absorber path and seatback-rotation control path are physically distinct, so reducing rotation does not simply mean making the entire seat rigid.
- A defined relationship between occupant trajectory envelope, seat travel, seatback angle, absorber work, and load-path continuity, with those quantities measured as explicit CAE/test observables.

3. SOURCE-BASED TECHNICAL FOUNDATION

The uploaded NHTSA/SAE materials provide the engineering basis for treating seat behavior as a coupled occupant-seat problem. The 2023 THOR-05F crash series used an instrumented advanced female ATD and documented complex thorax loading, variable lower-limb motion, and a formal seating procedure. The 2025 NHTSA/SAE work compared THOR-50M, BioRID-II and PMHS responses and evaluated seatback rotation and occupant kinematics. The accompanying virtual-testing study used a validated Honda Accord seat FE model to study seat collapse and strengthening under scaled rear-impact pulses.

Source	What it contributes	Status
THOR-05F study	Advanced ATD, 113 channels per dummy; seating target included pelvis ~20° ±2.5° and head 0° ±1.0°; thorax and lower-limb durability explicitly examined.	Source
THOR/PMHS 2025	Sled tests compared seatback rotation and head excursion; small sample size and sled-vs-full-vehicle limitations were explicitly acknowledged.	Source
Strengthened-seat 2025	Scaled 25.6 g pulse produced collapse-like kinematics at 47.5°; strengthened model reduced rotation to 24° and reduced modeled injury metrics in that simulation.	Source
TP-214D-09	Defines side-impact laboratory procedure and instrumentation discipline; explicitly states OVSC laboratory procedures are not regulations or NHTSA interpretations.	Source
Stronge impact mechanics	Impulse, force-time history, compliance, friction, contact and energy dissipation are central variables in impact mechanics.	Source / theory

Source-derived numbers should not be copied into claims as performance promises. For example, the 24° value belongs to a specific 2025 FE study of a Honda Accord seat model; it is not an approved T-OCS limit. Likewise, the 25.6 g scaled pulse is a comparative research input, not a certification test.

FIG. 1 - T-OCS concept / effective-force architecture

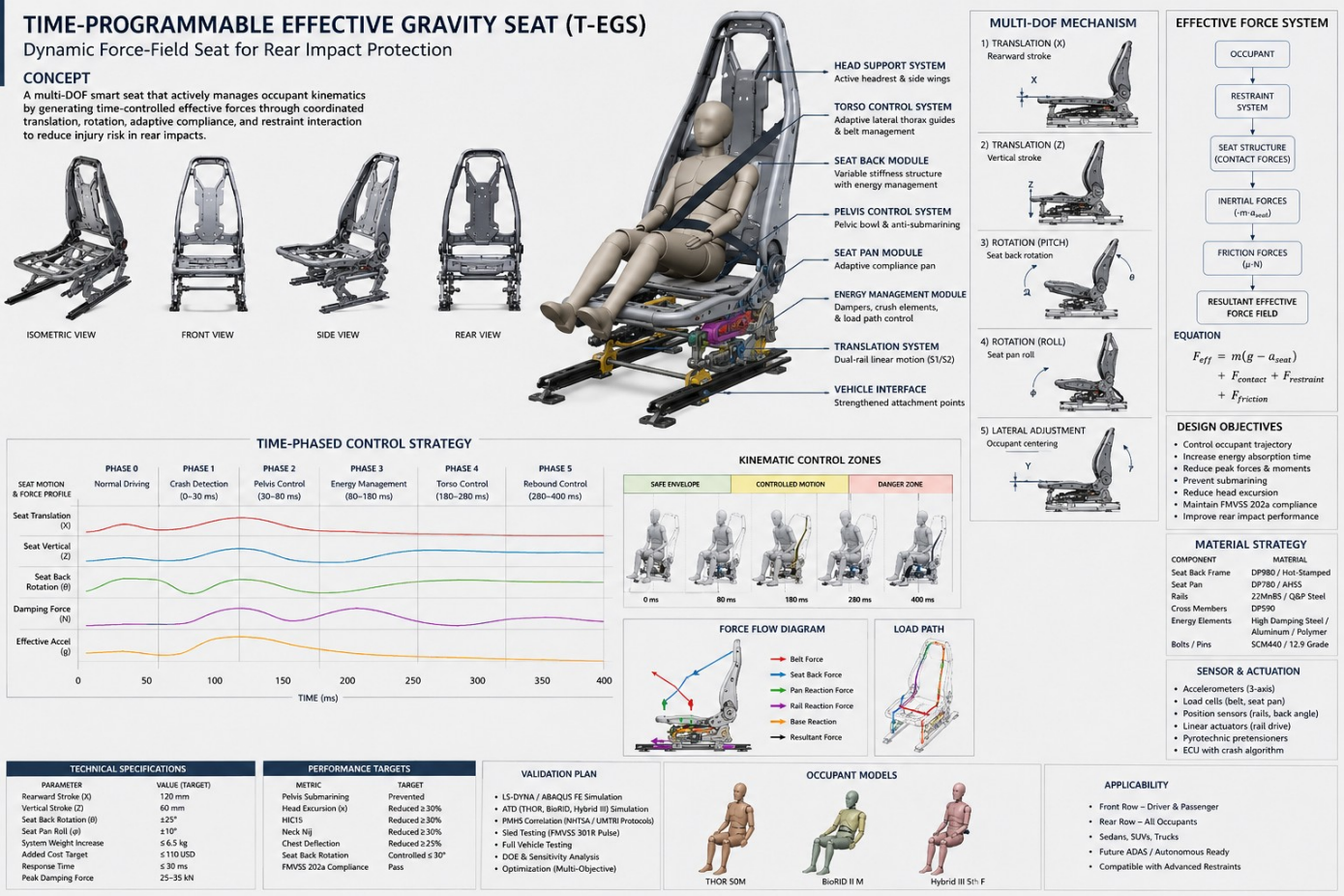
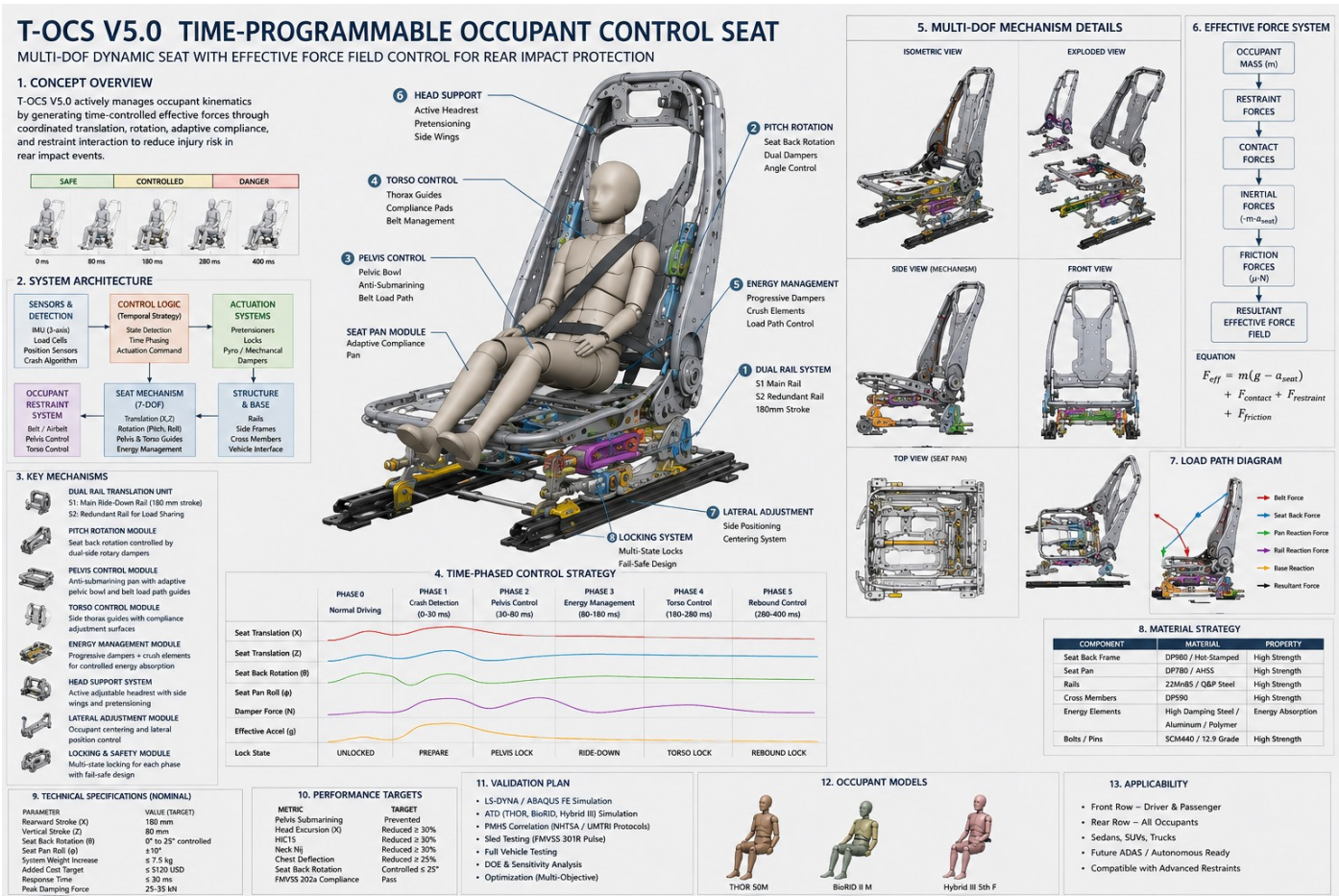


FIG. 2 - T-OCS V5.0 system architecture



Expanded V5 architecture: seat mechanism, dual rails, torso/pelvis control, energy management, locking logic and the inertial-frame effective-force equation.

FIG. 3 - Bilingual detailed CAD and industrial-development board



Exploded view and industrial-development concept used to communicate component hierarchy, rear-seat applicability and design intent. Dimensions shown in these concept boards remain nominal unless independently verified.

FIG. 4 - Detailed CAD engineering / manufacturing concept



FIG. 5 - Industrial rear-seat design concept

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TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT
Exploded View & Mechanism Details

A. EXPLODED VIEW - DETAILED CAD

- 1 Head Support Frame
- 2 Head Support Actuator (Linear)
- 3 Torso Side Guide (Left)
- 4 Torso Side Guide (Right)
- 5 Upper Torso Cross Member
- 6 Back Panel (Compliance)
- 7 Back Recline Hinge (Left)
- 8 Back Recline Hinge (Right)
- 9 Pelvis Bowl (Anti-Submerging)
- 10 Pelvis Adjust Actuator (Left)
- 11 Pelvis Adjust Actuator (Right)
- 12 Seat Pan (Adaptive Compliance)
- 13 Pan Front Hinge (Left)
- 14 Pan Front Hinge (Right)
- 15 Dual Rail System – S1 (Main)
- 16 Dual Rail System – S2 (Redundant)
- 17 Translation Actuator (S1/S2)
- 18 Energy Absorber – Damper
- 19 Energy Absorber – Crush Element
- 20 Base Frame (High Strength Steel)
- 21 Lateral Adjustment Unit (Left)
- 22 Lateral Adjustment Unit (Right)
- 23 Locking Mechanism (Multi-State)

B. MECHANISM DETAILS & JOINT DETAILS

B1 مفصل دوران الهيكل (إيسل)

B2 المسكّنات

B3 مفصل تعديل الجوانبي

B4 مفصل ضبط الوضع الجانبي

B5 مفصل القفل متعدد الحالات

B6 مفصل الطاقة - دايمر

B7 حركة قفل الجوانبي والانتعاش - عنصر سحب (ريد إيمرغنس) (180~400 mm)

D. DIMENSIONS (mm)

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INDUSTRIAL DESIGN + جاهز للتصنيع + تصميم صناعي واقعي

A. REAR SEAT – 2+1 CONFIGURATION

B. REAR SEAT – 3-SEAT CONFIGURATION

C. DESIGN FEATURES

راحة وأمان أنفث	تكيف تلقائي مع التضامم	تكامّل الأجزاء	مواد عالية الجودة	تنظيف فاحش
دعامات جانبية متحركة / داعم فطري للأنفث	تحكم زمني بالحرارة / وانضمام الطاقة	أجزاء مدمجة وخيار 4 / نقاط لتسهيل التجميع	قنوات على الطاولة ومسانير / وعناصر انضمام طاقة	جل طبيعي، وجهاطة / فاحشة

D. DIMENSIONS & PACKAGE

E. COLORS & FINISHES

F. APPLICATIONS

E. JOINT DEGREES OF FREEDOM

Joint ID	الوصف	DOF	النطاق
J1	دوران الهيكل (Roll)	1	-10° ~ +25°
J2	انحناء الجانب (Ball)	1	-5° ~ +5°
J3	ارتفاع الجذع (Z)	1	0 ~ 180 mm
J4	انحراف الجذع (X)	1	0 ~ 60 mm
J5	ضبط الوضع الجانبي (Y)	1	+30 mm
J6	تعديل الجوانبي (Rotation)	1	-15° ~ +15°

F. MATERIALS

Part Category	Material	Property
Seat Back Frame	DP980 / QAP-Steel	High Strength
Seat Pan	DP780 / AHSS	High Strength
Rails	Z2N/B5 / QAP-Steel	High Strength
Cross Members	DP950	High Strength
Energy Elements	High Damping Steel / AI	Energy Absorption
Bolts / Pins	SAC440 / 12.9 Grade	High Strength

G. TECHNICAL SPECIFICATIONS (NOMINAL)

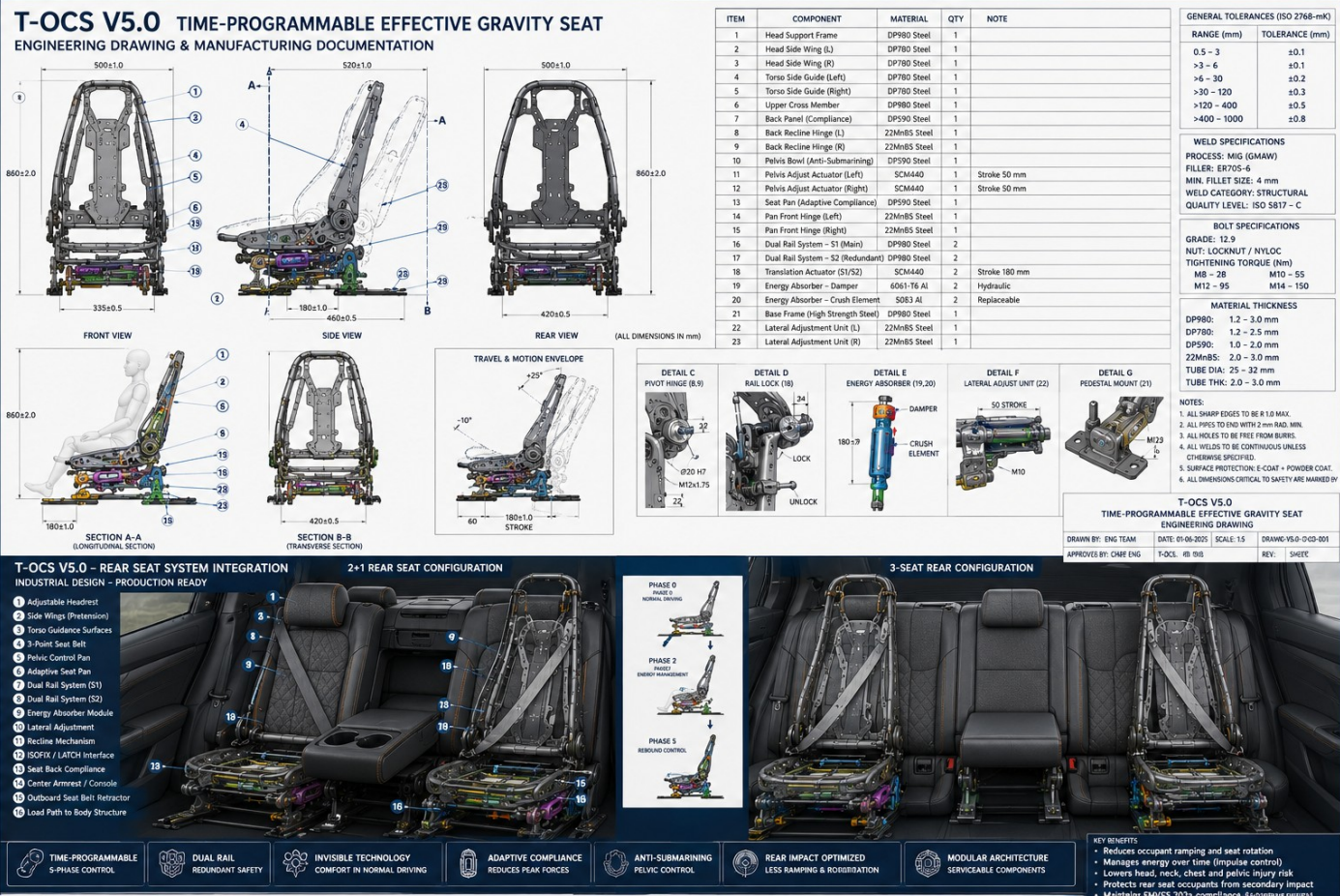
Parameter	Value / المواصفات
Reinforced Stroke	135 s
Vertical Stroke	80 mm
Seat Back Rotation (θ)	-10° ~ +25°
Seat Pan Roll (φ)	±5°
System Weight Increase	≤ 7.5 kg
Added Cost Target	≤ 120 USD
Response Time	≤ 30 ms
Peak Damping Force	25–35 kN

H. VALIDATION PLAN

- LS-DYNA / ABAQUS FE Simulation
- ATD (THOR, BioRID, Hybrid II)
- PMBH Correlation (BHTHS / UMTRI)
- Sled Testing (FMVSS 301R Pulse)
- Full Vehicle Testing
- DOE & Sensitivity Analysis
- Optimization (Multi-Objective)

Front/rear industrial design treatment illustrating a modular seat platform and rear-seat 2+1 / 3-seat configurations.

FIG. 6 - Engineering drawing and vehicle-integration board



Dimensioned engineering-style board including section views and rear-seat system integration. Treated as design-intent material, not manufacturing release.

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ENGINEERING DRAWING – MANUFACTURING PACKAGE

TIME-PARAMMABLE EFFECTIVE GRAVITY SEAT

UNITS: mm [inch]

SECTION A-A (LONGITUDINAL) SCALE 1:5

SECTION B-B (TRANSVERSE) SCALE 1:5

BILL OF MATERIALS			
No.	PART NAME	MATERIAL	QTY
1	Head Support Frame	DP980 Steel	1
2	Head Side Wing (L)	DP780 Steel	1
3	Head Side Wing (R)	DP780 Steel	1
4	Top Side Guide (L)	DP930 Steel	1
5	Top Side Guide (R)	DP930 Steel	1
6	Upper Torso Cross Member	DP980 Steel	1
7	Back Panel (Compliance)	DP930 Steel	1
8	Back Restline Hinge (L)	SCM440 Steel	1
9	Back Restline Hinge (R)	SCM440 Steel	1
10	Pelvis Bowl (Anti-Submerging)	DP780 Steel	1
11	Pelvis Adjust Actuator (L)	Steel + Motor	1
12	Pelvis Adjust Actuator (R)	Steel + Motor	1
13	Seat Pan (Adaptive Compliance)	DP930 Steel	1
14	Pan Front Hinge (L)	SCM440 Steel	1
15	Pan Front Hinge (R)	SCM440 Steel	1
16	Dual Rail System - S1 (Main)	22MnB5 Steel	1
17	Dual Rail System - S2 (Redundant)	22MnB5 Steel	2
18	Translation Actuator (SV/S2)	Steel + Motor	2
19	Energy Absorber - Damper	High Damp. Steel	2
20	Energy Absorber - Crush Element	Al 6061-T6	2
21	Base Frame (High-Strength Steel)	DP980 Steel	1
22	Lateral Adjustment Unit (L)	Steel	1
23	Lateral Adjustment Unit (R)	Steel	1

DETAILS – JOINTS & LOCKING

① RECLINE HINGE (8,9) SCALE 2:1

M12 BOLT CLASS 10.9
TORQUE: 95 Nm

② PAN FRONT HINGE (14,15) SCALE 2:1

M10 BOLT CLASS 10.9
TORQUE: 95 Nm

③ DUAL RAIL LOCK (18) SCALE 2:1

LOCK PIN Ø10 H7 HARDENED
SPRING

④ LATERAL ADJUSTMENT (22,23) SCALE 2:1

M8 BOLT CLASS 8.8
SLOT ADJUSTMENT ±25 mm

⑤ GUIDE RAIL SECTION (16,17) SCALE 1:1

3x1 [1 IN.]
60 [2.36]
2.8 [0.13]

⑥ ENERGY ABSORBER (19) SCALE 1:1

Ø45 [1.77]
STROKE 100 [4.72]

⑦ CRUSH ELEMENT (20) SCALE 1:1

Ø40 [1.57]
L 100 [3.94]

GENERAL TOLERANCES (UNLESS SPECIFIED)

DIMENSIONS : ±0.5
ANGULAR : ±0.5°
HOLE DIA : ±0.1
PROFILE : ±0.5

WELDING

FILLET WELD SIZE 6 mm (MIN.)
WELD CLASS : ISO 5817 / 10.9
WELD MATERIAL : ER70S-6

T-OCS V5.0 – REAR SEAT SYSTEM IN VEHICLE (INDUSTRIAL DESIGN – PRODUCTION READY)

2+1 REAR SEAT CONFIGURATION

3-SEAT REAR SEAT CONFIGURATION

KEY FEATURES – INTEGRATED

- ⌚ TIME-PARAMMABLE RIDE-DOWN
Dual rail translation with 100 mm stroke
- 🛡️ ADAPTIVE PELVIS CONTROL
Anti-submerging bowl + actuators
- 💧 PROGRESSIVE ENERGY ABSORPTION
Dampers + crush elements
- 👤 TORSO & HEAD GUIDANCE
Lateral thorax guides + active headrest
- 🔄 LATERAL ADJUSTMENT
Centering & optional positioning
- 🔒 MULTI-STATE LOCKING
Fail-safe mechanical locks
- 🔧 MODULAR SERVICEABILITY
Easy replaceable modules

T-OCS STRUCTURE (TRIM REMOVED)

SYSTEM SPECIFICATIONS (NOMINAL)

Total Weight: 29.8 kg ±1.0 kg
180 mm Ride-Down Stroke (X)
Vertical Stroke (Z): 60 mm
Seat Back Rotation (θ): ±25° (Controlled)
Seat Pan Roll (φ): ±10°
Operating Temp.: -40°C to +80°C
Dynamic Load Capacity: > 15 kN
Life Cycle: 15 Years / 200k Cycles

MATERIAL SUMMARY

High Strength Steel: DP980 / DP780 / DP930
Rails: 22MnB5 / Q&P Steel
Hinges & Pins: SCM440 (H)
Aluminum Parts: 6061-T6
Fasteners: Class 8.8 / 10.9
Dampers: High Damping Steel
Crush Elements: Aluminum Honeycomb

INDUSTRY STANDARDS / VALIDATION

- ✓ FMVSS 2024 Compliance (Head Restraints)
- ✓ FMVSS 207 Integrity (Seat Back Strength)
- ✓ FMVSS 2018 Pulse (Rear Impact)
- ✓ LS-DYNA / ABAQUS FE Simulation Protocols
- ✓ PMHS / ATD Correlation Full Vehicle Ready
- ✓ Sled Testing

AVAILABLE FINISHES

Black Leather	Dark Brown	Grey Leather	Beige Leather
Alicanta	Perforated Leather	Carbon Style	Brushed Metal

APPLICATIONS

- 🚗 Sedans
- 🚐 SUVs / Crossovers
- 🚌 MPVs / Minivans
- 🚘 Autonomous Vehicles
- 🏎️ Premium / Luxury
- 🚀 Future Mobility Platforms

NOTES

- Design prototype – not certified.
- Research intent: reduce injury risk in rear impacts by controlling occupant kinematics through time-parammable effective force field.
- All dimensions in mm [inch].
- Do not use for production without validation.

T-OCS V5.0 | Patent Draft / Invention Disclosure | NON-LEGAL DRAFT

FIG. 8 - Crash load-path analysis, 0-400 ms

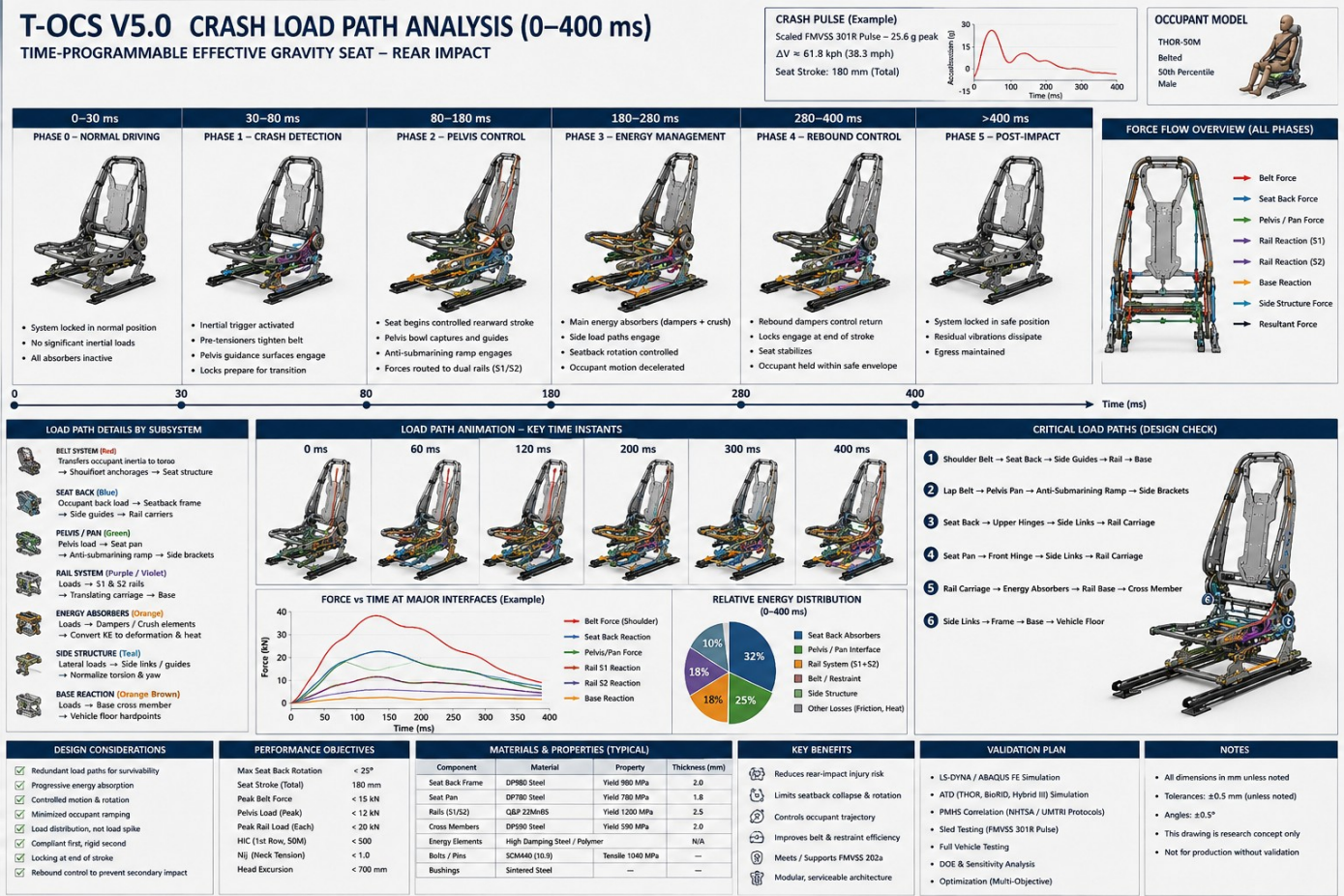
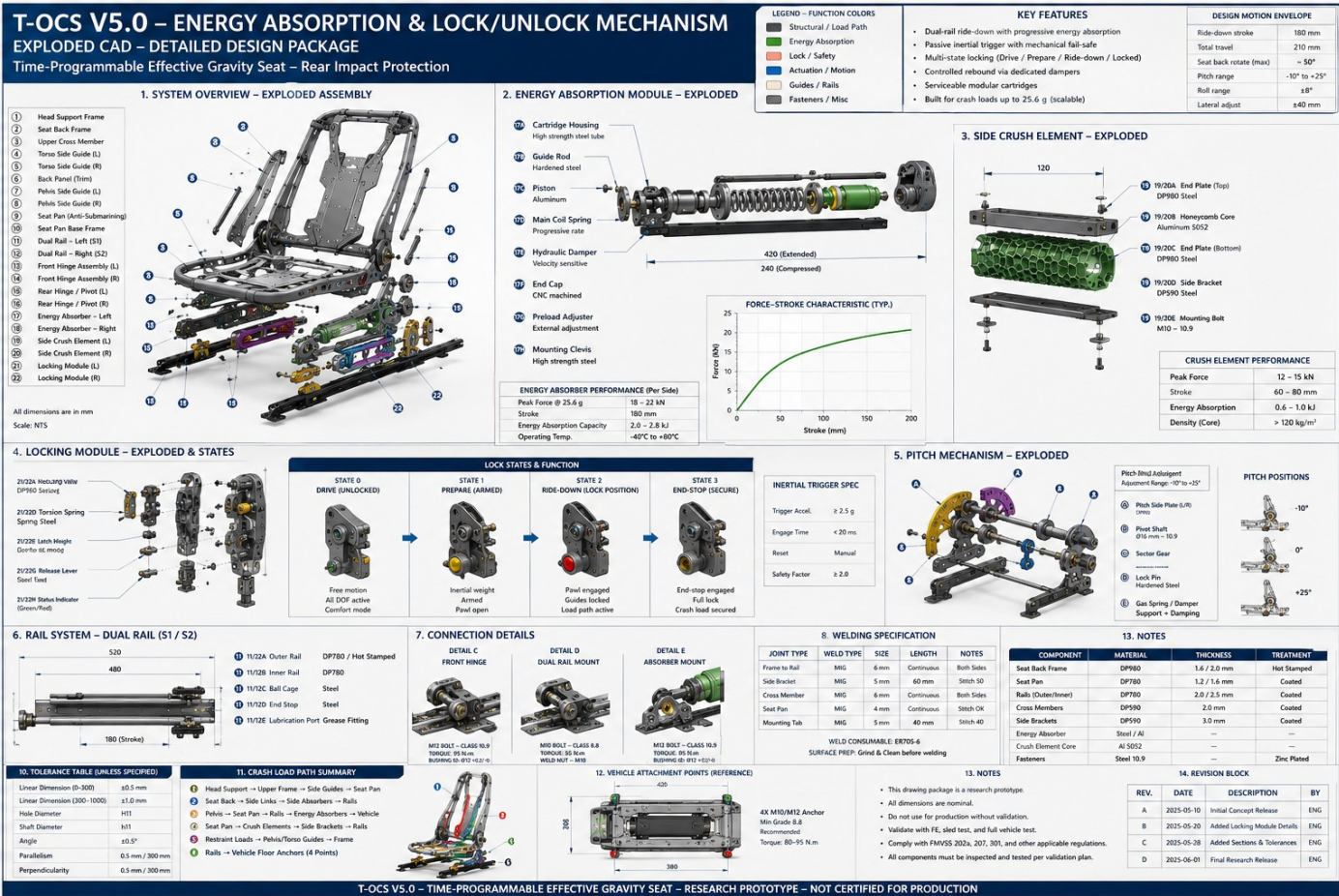


FIG. 9 - Energy absorption / Lock-Unlock mechanism



13. DETAILED DESCRIPTION OF THE INVENTION

13.1 System architecture. The T-OCS system comprises a vehicle interface/base, at least two longitudinal load paths, a seat pan, a seatback frame, one or more restraint interfaces, one or more energy management elements, and a locking/state-transition mechanism. The architecture is configured so that seat translation and seatback rotation can be controlled by distinct but coordinated mechanical paths.

13.2 Time-programmed states. A representative sequence is: S0 normal operation; S1 event recognition/arming; S2 controlled occupant capture and pelvis stabilization; S3 ride-down/energy dissipation; S4 rotation/rebound management; S5 secure post-event state. The sequence can be implemented mechanically, by sensing/actuation, or by a hybrid arrangement. The essential relationship is that the mechanical state of the seat changes as a function of event timing and/or detected dynamic state.

13.3 Effective-force interpretation. In a non-inertial seat frame, a commanded or passive seat acceleration contributes an inertial term. A useful engineering interpretation is $g_{eff}(t) = g - a_{seat}(t)$. The system does not create gravitational mass or a local gravitational field; it shapes contact and inertial force history.

13.4 Ride-down. The seat carriage may translate along dual rails through one or more energy absorbing elements. The energy absorber may use hydraulic, pneumatic, viscoelastic, metallic crush, frictional, or hybrid behavior, provided the force-stroke relationship is characterized before system validation.

13.5 Seatback rotation control. A dedicated rotation-limiting path may include bilateral links, dampers, hard stops, torsion elements or locking interfaces. The design intent is to avoid a false trade-off in which the seat is simply made rigid everywhere. The source FE study supports treating seatback rotation as a key design variable, but does not establish a universal target for this invention.

13.6 Occupant trajectory management. The system controls the trajectories of pelvis, torso and head by coordinating the seat pan, seatback, restraint geometry and ride-down motion. Design observables include seat travel, seatback rotation, head excursion, pelvis translation/rotation and restraint load history.

13.7 Redundant load paths. The preferred configuration uses left and right structural/energy paths so that a single local failure is not the sole route by which seat loads reach the vehicle body. Each path is intended to be independently measurable in testing.

13.8 Rebound control. A return damper, directional hydraulic valve, friction element, or secondary lock may limit reverse motion after the primary ride-down stroke. This state is intentionally treated separately because secondary interior contact can occur during rebound.

13.9 Fail-safe behavior. The primary occupant restraint path remains available even if the optional ride-down mechanism fails to arm. The design should default to a safe mechanical state when the trigger path is unavailable.

14. COMPONENT-BY-COMPONENT ENGINEERING DISCLOSURE

1. Vehicle interface / base frame - Carries the seat loads into vehicle hardpoints. Must be redesigned around actual vehicle package geometry; current dimensions are concept assumptions.
2. Dual rail S1/S2 - Provides guided translation and redundant load paths. Each rail should be instrumentable or independently modelable.
3. Rail carriage - Connects seat structure to the rails and transmits load into the absorber/lock modules.
4. Ride-down energy absorber - Controls seat travel and converts kinetic energy into controlled dissipative work. Force-stroke and hysteresis must be characterized.
5. Side crush / progressive absorber - Optional secondary energy path for lateral and local load management; material curve must be measured, not assumed.
6. Seat pan / pelvic bowl - Controls pelvis location and anti-submarining behavior; geometry must distribute load rather than form a rigid point contact.
7. Seatback frame - Primary torso/head support and load path; must resist unintended collapse and manage controlled rotation.
8. Seatback hinge / rotation limiter - Defines allowed pitch and transfers torque into the side structures. It may include friction, torsion, cam or damper elements.
9. Locking module - Multi-state mechanism for drive, prepare, ride-down and end-stop/rebound conditions.
10. Inertial trigger - Passive or hybrid trigger used to arm/release a state transition; requires fail-safe analysis.
11. Restraint interfaces - Seat-belt hardpoints and/or integrated guidance surfaces. Actual anchorage requirements are vehicle/regulation specific.
12. Torso guides - Provide distributed contact and trajectory guidance; should not create concentrated chest loads.
13. Head support - Controls head excursion and rebound clearance. The headrest and seatback must be validated together.
14. Sensor interfaces - Accelerometers, seat-position, seat-rotation and load-cell channels for state recognition and validation.

15. FRONT-SEAT AND REAR-SEAT EMBODIMENTS

Front seat embodiment. The same architecture can be packaged under a driver or front-passenger seat, with careful treatment of steering-column, pedal and dashboard clearance. The THOR-05F program demonstrates why lower-extremity and thorax loading cannot be reduced to a single scalar criterion; the source study reports complex and variable loading and uses extensive instrumentation. [cite not inserted in PDF]

Rear seat 2+1 embodiment. Each outboard seat module can have a dedicated T-OCS carriage and absorber path, while the center module uses a shared or narrower architecture. Packaging must preserve center-console/egress volumes and provide independent belt anchorage strategy.

Rear seat 3-seat embodiment. Three independent modules may be used where package width permits. Alternative architectures can share a common structural cross-member while preserving independent absorber/locking paths. The claim set should not rely on a particular number of seats.

Why the rear-seat context matters. The 2025 NHTSA/SAE work explicitly studied how seatback rotation and occupant excursion vary across vehicles and dummies, while acknowledging that small samples and sled-vs-full-vehicle differences limit direct injury conclusions.

Patent drafting principle. Describe the invention as a family of architectures capable of use in front and rear seating, but claim the mechanism generically enough that the inventive concept is not trapped by a single packaging embodiment.

16. VALIDATION AND TEST PROGRAM

Gate	Method	Pass/Fail evidence
V0 CAD integrity	Assembly interference, travel, joint range, packaging	No unintended interference; all stated DOF verified in CAD.
V1 reduced-order model	Multibody dynamics / sensitivity	Stable trajectories; bounded rotation and stroke for test pulses.
V2 component absorber	Force-stroke / hysteresis / temperature	Measured curve matches design corridor across repeated tests.
V3 lock/trigger	Bench trigger + repeated cycling + fault cases	Positive state transition; no unintended release; safe fallback.
V4 sled test	ATD + high-speed video + load cells	Occupant kinematics and seat loads within predefined engineering limits.
V5 HBM/FE	Validated seat + restraint + HBM/ATD model	No unphysical contact; stable correlation with sled data.
V6 full vehicle	Vehicle crash testing	Vehicle-level compliance and safety evidence.

Required measurements. Seatback rotation, seat travel, seat twist, head excursion, pelvis travel/rotation, restraint load histories, seat reaction forces, plastic strain/buckling/joint loads, absorber work $W = \int F(x)dx$, contact timing and friction state, and validated ATD/HBM injury metrics.

The uploaded TP-214D-09 emphasizes instrumentation calibration and traceability and states that its OVSC laboratory test procedures are not themselves regulations or NHTSA interpretations. The patent draft therefore treats such documents as methodological references and test-mapping sources, not as proof of compliance.

17. CANDIDATE CLAIM SET - LAWYER REVIEW REQUIRED

The following are candidate technical claims for discussion with a patent professional. They are intentionally narrower than a generic “energy absorbing seat” claim because the current Google Patents landscape shows substantial overlap with broad rail/absorber/inertial-lock architectures.

1. A vehicle seat system comprising: a seat structure having a seat pan and seatback; at least two longitudinally spaced load paths coupled to a vehicle structure; an energy-management assembly configured to permit controlled longitudinal displacement of the seat structure relative to the vehicle structure; and a multi-state mechanism configured to change the mechanical state of the seat system in response to a detected or predetermined dynamic event, wherein the longitudinal displacement is governed by a time-dependent sequence of states.
2. The system of claim 1, wherein a seatback-rotation control path is mechanically distinct from the longitudinal energy-absorption path such that a first mechanical element limits seatback rotation while a second mechanical element absorbs energy through controlled seat displacement.
3. The system of claim 1, wherein left and right load paths are configured so that each transfers a portion of an occupant-induced crash load to the vehicle structure through a respective absorber or structural path.
4. The system of claim 1, wherein the multi-state mechanism comprises a normal-driving state, an armed state, a ride-down state, and a rebound-control state.
5. The system of claim 4, wherein transition into the ride-down state permits a controlled seat stroke while maintaining an occupant restraint path independently of the energy-management assembly.
6. The system of claim 4, wherein the rebound-control state applies a directional damping or locking function to limit reverse seat movement after a primary ride-down stroke.
7. The system of claim 1, further comprising a pelvic geometry configured to resist forward pelvic migration while distributing pelvic contact load over a seat-pan region.
8. The system of claim 1, further comprising a sensor and controller or passive inertial trigger configured to initiate at least one state transition based on vehicle acceleration, seat acceleration, seat position, seatback rotation, or a combination thereof.
9. The system of claim 1, wherein the energy-management assembly comprises a replaceable cartridge having a characterized force-stroke relationship and being positioned in a load path between the seat structure and the vehicle structure.
10. The system of claim 1, wherein the seat system is configurable as a front-row seat module or a rear-row seat module without changing the fundamental time-dependent state logic.
11. A method of managing occupant motion in a vehicle seat comprising: maintaining a first mechanical seat state during normal driving; detecting a dynamic event; changing the seat to an armed or capture state; permitting controlled seat translation through an energy-management element; limiting seatback rotation independently of said translation; controlling rebound after the translation; and returning the seat to a secured state or a service state.
12. The method of claim 11, further comprising measuring seat translation and seatback rotation as independent variables and controlling the state transition based on one or both variables.
13. The method of claim 11, further comprising controlling pelvic migration using an anti-submarining seat-pan geometry while distributing load through bilateral load paths.
14. The method of claim 11, wherein the effective inertial acceleration experienced in a seat frame is treated as a function of seat acceleration according to $g_{eff}(t) = g - a_{seat}(t)$, without generating a local gravitational field.
15. A modular energy-management cartridge for a vehicle seat comprising a guided rod or piston, a damping element, a progressive resistance element, first and second connection interfaces, and a serviceable housing configured for insertion into a vehicle-seat load path, wherein the cartridge provides a predetermined or characterized force-stroke response over at least a first ride-down region and a second end-control region.

18. CLAIM STRATEGY, ALTERNATIVE EMBODIMENTS, AND ABSTRACT

Independent-claim strategy. Keep the independent apparatus claim centered on the interaction between time-dependent state transitions, dual load paths, controlled translation and separately controlled rotation. Put specific hardware (cam/pawl, hydraulic cartridge, crush element, sensor/actuator type) into dependent claims unless the final prior-art search shows a specific mechanism to be the strongest novel feature.

Alternative embodiments. The energy absorber may be hydraulic, pneumatic, frictional, viscoelastic, metallic crush, composite crush, or hybrid. The trigger may be passive inertial, electromechanical, pyrotechnic, software-controlled, or hybrid. The lock may use pawl/cam, wedge, ratchet, pin, dog clutch, or equivalent geometry. Such alternatives should be described to preserve claim breadth while the claims remain anchored to the time-dependent functional relationships.

Abstract (draft). A vehicle seat system includes a seat structure coupled to a vehicle through at least two load paths and a time-dependent occupant-control mechanism. The mechanism transitions between normal, armed, controlled ride-down, rebound-control and secured states in response to vehicle or seat dynamics. During a crash event, the seat structure can translate through a characterized energy-management assembly while a distinct path limits seatback rotation. Pelvic geometry, restraint interfaces and distributed side structures cooperate to manage occupant trajectory and to distribute load into the vehicle structure. The system uses controlled inertial and contact forces rather than generating a local gravitational field. The architecture can be implemented in front or rear vehicle seating, including modular two-plus-one and three-seat rear-seat configurations.

19. THOR-05F AS A RESEARCH AND VALIDATION MODEL - NOT AS OWN TECHNOLOGY

The 2023 THOR-05F study is included in this package because it demonstrates how a professional crash-safety research program defines the test article, instrumentation, seating posture, durability, and post-test inspection. It reports 113 channels per THOR-05F, repeated qualification checks, explicit seating targets, and component repair/requalification. [filecite\turn8file3\L1-L6]

The same study reports thorax deflection ranges that varied by vehicle and occupant position and notes that the location and rate/distribution of loading matter in addition to peak deflection. It also documents highly dynamic ankle motion across three rotational degrees of freedom. [filecite\turn8file4\L1-L4]

The 2025 NHTSA/SAE work explicitly compares THOR-50M, BioRID-II and PMHS responses and measures seatback rotation and head excursion, while warning that small sample size and sled-vs-full-vehicle differences limit direct injury conclusions. [filecite\turn8file5\L1-L6]

The THOR-05F topic is also current research infrastructure rather than a patent source for the T-OCS. NHTSA's current materials state that THOR-05F development emphasizes biofidelity, durability, repeatability/reproducibility and documentation, and that design specifications have been released for automakers to use in testing.

[cite\turn948651search24\turn948651search1]

Therefore: THOR-05F methods inform how T-OCS should be validated, but the dummy itself, its measurements, and NHTSA's proprietary/official design information are not being claimed as part of the invention.

20. REFERENCES AND SOURCE MAP

Source	Document	Use	Link
NHTSA/SAE 2023	Evaluation of THOR-05F in a Full Vehicle Crash Series	Uploaded PDF: 15874-SAE GIM 2023 THOR05F for distrib_032223-tag_.pdf; also publicly available at NHTSA.	https://www.nhtsa.gov/sites/nhtsa.gov/files/2023-03/15874-SAE%20GIM%202023%20THOR05F%20for%20distrib_032223-tag_1.pdf
NHTSA/SAE 2025	A Preliminary Biofidelity Assessment of THOR and BioRID in a Moderate-Speed Rear Impact Sled Test	Uploaded PDF: 16616-Tesny_SAE 2025 GI Prelimin Biofidelity Asses_2.pdf	https://www.nhtsa.gov/
NHTSA/SAE 2025	Seat Assessment in Standardized Rear Impact Virtual Testing	Uploaded PDF: 16616-Greb_SAE 2025 Seat Assess In Standard Rear I_260902_003053.pdf	https://www.nhtsa.gov/
NHTSA	TP-214D-09 Laboratory Test Procedure for FMVSS No. 214	Uploaded PDF: tp214d-09_.pdf	https://www.nhtsa.gov/
Stronge	Impact Mechanics	Uploaded PDF supplied by applicant; used for mechanics framing (impulse, compliance, contact, friction, energy).	
Google Patents	US20200231073A1 / US11390198B2	Energy attenuating vehicle seat assembly.	https://patents.google.com/patent/US20200231073A1/en
Google Patents	EP3911536A1	Energy attenuating vehicle seat assembly.	https://patents.google.com/patent/EP3911536A1/en
Google Patents	WO2021197809A1	Seat rail pair for longitudinal adjustment and energy absorber and locking device.	https://patents.google.com/patent/WO2021197809A1/en
Google Patents	US4634182A	Seatback recliner mechanism and inertia operated lock.	https://patents.google.com/patent/US4634182A/en
Google Patents	US6050635A	Seat pan including anti-submarining device.	https://patents.google.com/patent/US6050635A/en
Google Patents	US5125598A	Pivoting energy attenuating seat.	https://patents.google.com/patent/US5125598/en

Web-search basis checked 2026-09-02. Google Patents records are used for prior-art screening, not as a legal opinion on validity or enforceability.

21. PRE-FILING CHECKLIST - DO NOT SKIP

- ☐ Document the earliest conception date, versions, drawings, notebooks, CAD history and any disclosure dates.
- ☐ Freeze the inventive concept into a single consistent vocabulary: seat carriage, dual load path, ride-down absorber, separate rotation-control path, multi-state lock, rebound state, occupant trajectory.
- ☐ Run a professional novelty + inventive-step search before filing. The preliminary Google Patents scan already shows meaningful overlap with rail/absorber/inertial-lock architectures.
- ☐ Do not publicly disclose a final patentable embodiment before a patent professional advises on filing strategy and applicable grace-period rules.
- ☐ Replace every concept dimension with either a vehicle-package value, a measured prototype value, or an explicitly labeled design assumption.
- ☐ Create controlled CAD revisions and a drawing index; every figure in the patent should map to stable reference numerals.
- ☐ Build a validation matrix linking each claim limitation to a testable physical or CAE observable.
- ☐ Separate measured results, simulation predictions, and target values in all investor and academic materials.

Ruthless conclusion

Your project is no longer just a visual seat concept, but it is also not yet a proven patent. The strongest path is to protect a specific system architecture and state-transition mechanism, then prove the mechanism experimentally. The most dangerous mistake now would be to file a broad, emotional claim around “artificial gravity” that the existing prior art can dismantle.